Preliminary Study: Contrastive Activation Addition for Natural Code-Switched Speech in Speech-LM TTS

cs-tts-steering project

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Abstract

We tested whether contrastive activation addition (CAA), following CoCoEmo (Wang et al., ICML 2026), can make Hindi–English code-switched TTS output sound more like natural speech. We built a steering vector $v = \bar{h}_{\text{natural}} - \bar{h}_{\text{synthetic}}$ from 76 training transcripts. Each transcript has a natural TV recording and default TTS output. We extracted the vector in three conditions: Indic-Speak with voice Amit, Indic-Speak with voice Kavya, and Orpheus Hindi with voice Ritika. A linear natural-vs-synthetic direction exists in every condition: the held-out projection AUC is ≈ 1.0, against a permutation 95th percentile of 0.62. Adding it to attention outputs at small strength ($\alpha^* = 1$) is causal: it moves the model's internal representation and an external natural-vs-TTS classifier in the intended direction, beyond random vectors of the same norm. However, it does *not* bring the output closer to the paired natural recording, in embedding similarity or in prosody. The acoustic changes are mostly a shift in global pitch level and pausing toward the TV speakers. All three conditions are therefore **no-go on the pre-registered objective criteria**. Indic-Kavya is the most promising: it is the only condition that passes all guardrails and shows a symmetric $\pm v$ effect. Indic-Speak is more steerable than Orpheus. Informal listening by the team does not follow the objective ranking: steered Indic-Amit sounds *different* but not clearly more natural, while steered Orpheus-Ritika sounds somewhat *more* natural. The formal blind test is pending. All results are limited by imperfect transcripts: the Devanagari/English inputs were derived from speech via a romanized transcription and then machine transliteration, losing information at each step.

1 Motivation and question

Speech-LM TTS systems read code-switched Hinglish intelligibly but sound read-aloud and unnatural. CoCoEmo shows that in hybrid TTS, emotional prosody is produced mainly by the speech language model, and that mean-difference vectors added to its activations control emotion without retraining. We ask whether the same mechanism works for a different attribute, naturalness: *does the LM contain a linear "natural vs. synthetic" direction, and does adding it make code-switched output more natural without hurting intelligibility or changing the voice?* This is a feasibility study with a go/no-go answer per condition. The full plan is in `docs/caa_pilot_plan.md`.

2 Setup

2.1 Data

We used 100 code-switched MaSaC utterances (`Listening_Examples_100`). Each has a natural TV-show recording (6 speakers, about 65% male, stereo 44.1 kHz) and default TTS output for the same mixed Devanagari/English transcript. The transcripts were produced in two lossy steps.

First, the speech was transcribed into romanized Hindi/English. Second, that romanized text was machine-transliterated into Devanagari for Hindi words, with English left in Latin script. Each step lost information (§4), and the final inputs contain spelling errors. We dropped five utterances whose generation hit the token cap (4 Orpheus, 1 Amit) from *all* conditions. The remaining 95 were split at the utterance level into 76 train (16 of which form a dev subset for tuning) and 19 test, with seed 20261029. All conditions share the split. The only audio processing was mixing to mono and resampling to 24 kHz.

2.2 Conditions

Condition	Model	Voice
Indic-Amit	Indic-Speak (`bodhan-ai/indic-speak`)	Amit (M)
Indic-Kavya	Indic-Speak	Kavya (F)
Orpheus-Ritika	Orpheus 3B Hindi FT (`canopylabs/3b-hi-ft-research_release`)	Ritika

Pinned revisions: Indic-Speak 76e0814f, Orpheus 70c4b32a. Both LMs are Llama-3.2-3B models (28 layers, $d = 3072$) that emit SNAC audio codes, 7 tokens per frame. Indic-Speak decodes them with a fine-tuned Vocos and Orpheus with the stock SNAC decoder. As in CoCoEmo, we steer only the LM.

2.3 Steering vector extraction

Indic-Speak and Orpheus have no reference-audio input, unlike CosyVoice2. We therefore **teacher-forced** each clip instead: the model's text prompt (with the condition's voice name) followed by the clip's SNAC tokens. Natural and synthetic wavs were both re-encoded with SNAC (rev. d73ad176), so the two classes go through the same encoder. Indic-Speak's rule of dropping consecutive frames with a repeated first code was reproduced, and round-trip tests against each model's own decoder function pass on real audio.

Within a condition, the natural and synthetic inputs differ only in the audio. We recorded the residual stream and the attention output (`o_proj`) at every layer, pooled two ways (last audio token and mean over audio tokens). That gives 112 sites per condition. For each site,

$$v^{(l,o)} = \frac{1}{|D_{\text{nat}}|} \sum_{i \in D_{\text{nat}}} h_i^{(l,o)} - \frac{1}{|D_{\text{syn}}|} \sum_{j \in D_{\text{syn}}} h_j^{(l,o)}.$$

At inference we add the vector and then rescale to the original norm: $\tilde{h} = h + \alpha v$, $\tilde{h} \leftarrow \tilde{h}\, \|h\| / \|\tilde{h}\|$. This is applied at every position of the selected sites.

2.4 Site and strength selection

All selection used only training utterances.

- **Diagnostics.** For each site we ran a regularized linear probe (5-fold CV by utterance), computed the held-out AUC of projecting onto v (with v estimated on 4 folds), and compared both against a label-permutation control.
- **Sweep 1.** A layer-wise steering sweep on the dev subset over the 6 shortlisted sites, $\alpha \in \{2, 4\}$.
- **Sweep 2.** Top-K combinations, $K \in \{1, 2, 3\}$, with $\alpha \in \{0.5, 1, 2, 3, 4, 6\}$.

- **Selection rule** (fixed in advance). A configuration is admissible if three guardrails hold: WER/CER at most the condition's own baseline +2 points, S-SIM ≥ 0.75, and failure rate at most $\max(5\%, \text{baseline failure rate})$. Among admissible configurations we chose the best composite of the three naturalness metrics. The mild strength is the next-lower admissible α.

2.5 Test protocol and metrics

We ran 19 test utterances $\times$ 3 seeds per condition under seven generation conditions: $\alpha = 0$; v at α^*; v at the mild strength; three random vectors of the same norm at α^*; and $-v$ at α^*. That is 1,197 clips. Following CoCoEmo, emotion classification was replaced by naturalness measures:

- **M1** (in place of TEP): log-odds of "natural" from a logistic classifier on WavLM-base-plus embeddings. It was trained on the 76 training utterances' natural clips vs. all three conditions' synthetic clips, and has held-out AUC = 1.0. We use the logit because $P(\text{natural})$ saturates.
- **M2** (in place of E-SIM): cosine similarity of WavLM embeddings to the *paired* natural clip.
- **M3**: prosody distance to the paired natural clip, lower is better. It is the IQR-scaled absolute difference averaged over 7 features: F0 median, SD and range; intensity SD; speaking rate; pause count; and pause time. We also report DTW-aligned F0 and energy contour CCC.
- **Guardrails**: WER/CER from Whisper-large-v3, used only relative to each baseline because even natural speech scores WER 0.70 against these transcripts; S-SIM, the WavLM-SV cosine to the unsteered clip with the same seed; and failure rate (no end-of-speech token, token budget reached, or abnormal duration).

Deltas are computed against $\alpha = 0$, averaged over seeds within each utterance, with 95% bootstrap CIs over utterances and paired Wilcoxon tests. The three conditions are compared with a Friedman test and pairwise Wilcoxon tests. Following the project's rules, we did not use UTMOS or SQUIM. Naturalness is ultimately judged by bilingual listeners (§7).

3 Results

3.1 A natural-vs-synthetic direction exists in every condition

Table 1: Diagnostics on the 76 training pairs (112 sites per condition). Permutation control: 200 pair-swaps for AUC, 3 per site for probes.

	Indic-Amit	Indic-Kavya	Orpheus-Ritika
Probe acc. = 1.0 (sites)	86/112	108/112	82/112
Probe acc. under permutation	0.48	0.49	0.49
Held-out projection AUC (min)	0.98	0.977	0.89
Permutation AUC, mean / max 95th pct.	0.48 / 0.62	0.48 / 0.62	0.49 / 0.62
Held-out Cohen's d (median, resid / attn)	8.7 / 7.7	7.9 / 7.2	5.7 / 5.8
$\|v\|/\|h\|$ (median, resid / attn)	0.38 / 0.75	0.31 / 0.70	0.28 / 0.73
Audio-token NLL, natural vs. synthetic (nats)	5.74 vs. 2.93	5.75 vs. 2.87	5.93 vs. 3.17

Separation is near-perfect at almost every site, including layer 0 (Table 1). This is what the plan expected, because speaker and recording channel make the classes trivially separable. As a result, probes and AUC cannot choose a site, and the steering sweep had to.

Indic-Speak separates most strongly in early residual layers (2–6) and again at 23–25. Orpheus's separation grows with depth. The Amit and Kavya vectors are closely aligned (cosine 0.8–0.9),

except at attention layers 16–18 (0.33–0.51). The direction is therefore largely shared across the two voices.

Natural audio has about twice the per-token NLL of synthetic audio, and v is partly aligned with the high-vs-low-NLL direction among synthetic clips (cosine up to 0.70). So v also encodes "audio the model finds surprising".

3.2 Selected configurations

Table 2: Selection on the dev subset. All selected sites are attention outputs (`o_proj`) with a mean-pooled v.

Condition	Admissible / tried	Layers	α^*	Mild α
Indic-Amit	8/18	9, 23	1.0	0.5
Indic-Kavya	12/18	9, 27, 23	1.0	0.5
Orpheus-Ritika	3/18	25, 23, 9	1.0	0.5

Residual-stream steering at $\alpha \geq 2$ broke generation in every condition: runaway outputs, missing end-of-speech, or Orpheus frames that could not be decoded. Only attention-output steering at small α was usable (Table 2).

In sweep 1, steering effect was *negatively* correlated with probe separation (-0.83, -0.39, -0.62), because the broken residual sites were also the best-separating ones. This is the opposite of CoCoEmo's finding that separability predicts steerability.

3.3 Test results

Table 3: Test results (19 utterances × 3 seeds). Δ is relative to $\alpha = 0$ with 95% bootstrap CI. M1, M2: higher is better. M3: lower is better. Bold: CI excludes 0 in the intended direction.

Cond.	Gen.	Fail	WER	CER	S-SIM	ΔM1	ΔM2	ΔM3
Amit	$\alpha = 0$	0.00	0.528	0.434	–	–	–	–
	v, α^*	0.04	0.741[†]	0.640[†]	0.936	**+2.20** [1.71, 2.61]	+0.015 [0.000, 0.030]	+0.046 [−0.02, 0.13]
	v, 0.5	0.00	0.537	0.435	0.969	**+0.77** [0.47, 1.07]	**+0.013** [0.004, 0.024]	+0.003 [−0.04, 0.05]
	$-v$, α^*	0.00	0.549	0.439	0.977	−0.16 [−0.42, 0.12]	−0.010 [−0.018, −0.002]	+0.062 [0.02, 0.11]
Kavya	$\alpha = 0$	0.00	0.488	0.408	–	–	–	–
	v, α^*	0.02	0.501	0.407	0.951	**+1.80** [1.18, 2.72]	+0.015 [−0.014, 0.042]	+0.054 [−0.000, 0.118]
	v, 0.5	0.00	0.482	0.397	0.972	**+0.43** [0.22, 0.63]	**+0.010** [0.001, 0.019]	+0.011 [−0.04, 0.05]
	$-v$, α^*	0.00	0.522	0.438	0.975	−0.40 [−0.68, −0.18]	−0.002 [−0.013, 0.008]	−0.024 [−0.06, 0.01]
Orpheus	$\alpha = 0$	0.00	0.615	0.519	–	–	–	–
	v, α^*	0.05	0.642	0.647	0.938	**+0.94** [0.48, 1.52]	+0.007 [−0.022, 0.034]	−0.004 [−0.08, 0.08]
	v, 0.5	0.04	0.671	0.675	0.931	+0.20 [−0.14, 0.59]	+0.006 [−0.02, 0.03]	+0.055 [−0.02, 0.16]
	$-v$, α^*	0.09	0.772	0.633	0.938	+0.29 [0.03, 0.59]	−0.001	−0.028

[†]One Whisper hallucination (utterance `train_1394_7`, seed 1235; output 12× the reference length). Without that clip, WER is 0.485 vs. 0.471, and the median clip WER is unchanged (0.476). The pre-registered corpus-level guardrail still counts as failed.

Against random vectors. These are per-utterance differences, v at α^* minus the mean of three random vectors of the same norm.

- **M1:** +2.09 [1.69, 2.45] for Amit, +1.76 [1.13, 2.64] for Kavya and +0.83 [0.40, 1.36] for Orpheus (all Wilcoxon $p < 0.001$). Random vectors alone move M1 by only −0.02 to +0.25.

- **M2:** no CI excludes 0. Kavya has Wilcoxon $p = 0.023$, but its CI is $[-0.011, 0.035]$.
- **M3:** Kavya is *worse* than random (+0.069 [0.025, 0.127], $p = 0.004$); Amit and Orpheus are not significant.

Mechanistic check. Each steered clip was re-encoded and teacher-forced through the *unsteered* model, then projected onto the unit v at the first selected site.

- v at α^* moves the projection by +1.48 (Amit), +1.07 (Kavya) and +8.4 (Orpheus). $-v$ moves it by −0.42, −0.53 and −4.4. Random vectors give about 0. All v and $-v$ effects have $p < 10^{-4}$.
- The natural–synthetic gaps along these directions are 4.7, 4.6 and 38.7, so $\alpha^* = 1$ closes about 20–30% of the gap.
- The intervention is causal and dose-dependent at the level of the representation.

What changes acoustically. Median F0 moves toward the natural pool's mean of 14.1 st (re 100 Hz):

- Amit: 7.9 → 12.5 st
- Kavya: 13.8 → 14.5 st
- Orpheus: 17.6 → 15.7 st

Speech also gets slightly slower, with more and longer pauses (+0.3 to +0.7 per clip). Contour agreement with the paired natural clip barely changes: F0 and energy CCC rise by only +0.01 to +0.09. In other words, the vector moves the global pitch level and pausing toward the TV speakers, not the utterance-specific delivery. This is the speaker/gender confound the plan anticipated. For male Amit it even *raises* pitch toward the mixed-gender pool.

3.4 Three-way comparison

On the same 19 utterances, the gains in M1 differ by condition (Friedman $\chi^2 = 12.7$, $p = 0.0017$): Amit ≥ Kavya > Orpheus. Amit vs. Orpheus has $p = 0.003$, Kavya vs. Orpheus $p = 0.016$, and Amit vs. Kavya $p = 0.066$. Gains in M2 ($p = 0.53$) and M3 ($p = 0.37$) do not differ. Indic-Speak is more steerable along v than Orpheus, and its two voices respond alike. Orpheus has the weakest baseline and the noisiest response, and it is the only condition where $-v$ did not reverse M1.

3.5 Informal listener feedback

Before the formal blind test, team members (Hindi–English bilinguals) listened to unsteered and steered clips. This feedback is verbal and informal: not blinded, no fixed protocol, and no ratings. It should be read as a pointer for the formal test, not as a result.

- **Indic-Amit:** steering makes the voice sound *different*, but not necessarily more naturalistic. This fits the acoustics: Amit has the largest pitch shift (+4.6 st on average, about +10 st in one of the five side-by-side examples) and the lowest S-SIM.
- **Orpheus-Ritika:** steering makes the output sound *a bit more naturalistic*. This is notable because Orpheus had the smallest objective gain (M1) and failed the guardrails. Its pitch shift went the other way (−1.9 st, toward the natural pool's mean) and it gained pauses.
- **Indic-Kavya:** no specific feedback yet.

Table 4: Pre-registered criteria (plan §8), judged on the test set.

Criterion	Indic-Amit	Indic-Kavya	Orpheus-Ritika
1. Direction exists (projection AUC $\gg$ permutation)	pass	pass	pass
2a. M1 improves vs. $\alpha = 0$ and random	pass	pass	pass
2b. M2 or M3 improves vs. $\alpha = 0$ and random	fail	fail (M3 worse)	fail
2c. $-v$ reverses the effect	partial	pass (M1)	fail
3. Guardrails on test	fail (WER/CER†)	pass	fail (CER, failures)
4. Listeners prefer steered (formal test)	pending	pending	pending
Informal team feedback (§3.5)	different, not more natural	–	somewhat more natural
Objective verdict	**no-go**	**no-go (best candidate)**	**no-go**

3.6 Go/no-go

4 Discussion

What worked. The contrast is strongly linearly encoded in both backbones. Small attention-output interventions are causal and dose-dependent, and they leave the voice largely intact (S-SIM ≥ 0.93) and intelligibility mostly intact. This reproduces CoCoEmo's core mechanism on two new backbones and a new attribute. The LM is a workable place to steer.

What did not. The vector does not encode naturalness *as defined by the paired recording.* Three confounds are present in v by construction:

1. speaker and gender identity of the TV actors;
2. the TV recording channel (stereo mix, music, laugh track);
3. "surprising" audio, shown by the twofold NLL gap.

The external classifier (M1) can be satisfied by moving toward any of these, which explains why M1 rises while M2 and M3 do not. Our result differs from CoCoEmo, where separability predicted steerability. There, the contrast was matched on speaker and transcript, so the direction isolated the target attribute. Here, separation is trivially high everywhere, and the most separable residual sites are the least steerable.

Objective metrics vs. listeners. The informal feedback runs against the objective ranking. Amit moved most on M1 but was heard as merely different. Orpheus moved least but was heard as somewhat more natural. A large shift in voice and pitch is easy for a natural-vs-TTS classifier to reward, but it is not what makes speech sound natural. A smaller shift in pitch level and pausing, as in Orpheus, may be. The paired-clip metrics (M2, M3) are also weakened by the transcript problem below. Together this suggests that our objective metrics are not yet good proxies for perceived naturalness. Orpheus should not be dropped on objective grounds alone, and the formal blind test carries more weight than the objective verdict.

Imperfect transcripts. The text used for both the TTS inputs and teacher forcing went through a lossy chain: speech $\rightarrow$ romanized Hindi/English transcript $\rightarrow$ machine-transliterated Devanagari/English. The romanized step loses what romanization cannot express, such as vowel length, retroflex and dental contrasts, nasalization, and hesitations or repairs that were not written down. The transliteration step guesses spellings from an ambiguous romanization and makes errors; it can also mislabel the language of a word. As a result:

- The "same transcript" pairing is only approximate. When natural audio is teacher-forced under a text that does not match what was actually said, part of the activation difference and

part of the twofold NLL gap may reflect this text–audio mismatch rather than naturalness. This adds a fourth confound to v.
- The TTS models were given inputs with spelling errors, which limits how natural even the steered output can sound.
- WER (0.70 even for natural speech) and the paired comparisons M2 and M3 are measured against imperfect references.

Implication. The bottleneck is the contrast data, not the mechanism. Cleaner contrasts and verified transcripts are the next step before scaling up.

5 Limitations and deviations from the plan

- The test set is small: 19 utterances, 3 seeds, and one random split.
- Transcripts are imperfect: speech was transcribed to romanized Hindi/English and then machine-transliterated to Devanagari/English, losing information at each step (§4). This affects the contrast itself, the TTS inputs, WER (0.70 even for natural speech), M2 and M3.
- The formal blind listening test is not done yet; listener evidence so far is informal team feedback (§3.5). The token-cap clips were excluded but not listened to.
- M1 uses the classifier logit, because $P(\text{natural})$ saturates.
- The failure guardrail was relaxed to $\max(5\%, \text{own baseline})$ because Orpheus's dev baseline failed 12–19%. It was applied the same way to all conditions.
- Sweep 1 ranked sites with a usability filter. The strict selection rule was applied unchanged in the final step.
- The composite score is the mean of standardized ΔM1, ΔM2 and $-\Delta$M3. The plan did not define it.
- Generation was batched, so a clip reproduces only with the same batch composition. Batch seeds are recorded per clip.
- A padding bug was found and fixed. Hugging Face's repetition penalty also scores padding tokens, and a stop-token pad suppressed end-of-speech in padded rows. The affected sweep was discarded and rerun.
- Not done: the "generated tokens only" injection ablation, and prosody at code-switch points (no word alignments available).

6 Next steps

1. **Blind listening** (§7). Run it first, to decide criterion 4 for all three conditions. The informal feedback makes Orpheus-Ritika worth testing despite its objective result, and Indic-Kavya has no listener feedback yet.
2. **Better transcripts.** Have bilingual annotators verify or correct the Devanagari/English inputs directly against the audio, then re-extract v. This tests how much of the contrast and the NLL gap comes from text–audio mismatch.
3. **Cleaner contrasts.** Options:
   - natural vs. natural speech from the *same* speaker in different speaking styles;
   - speaker-matched natural data, for example natural recordings of the Kavya/Amit voice artists if available;
   - synthetic-vs-synthetic controls, to estimate how much of v is channel or NLL.

4. **Remove known confounds from v.** Project out the speaker/F0 direction and the high-NLL direction, then re-test M2 and M3.
5. **Remaining ablations.** Steer generated tokens only, and add code-switch-point prosody once word alignments exist.

7 Listening package

The package has 63 blinded A/B items:

- 57 pairs of $\alpha = 0$ vs. v at α^* (19 utterances $\times$ 3 conditions);
- 6 random-vector control pairs;
- an optional natural anchor clip per item (a different speaker).

The A/B order is randomized and the items are shuffled across conditions. The package contains audio and transcripts, so it is kept on kedi only. The unblinding key must not be given to raters.

A Provenance

- **Run:** `/local/speech/users/sp4050/outputs/caa_pilot/20261029-1320_listening100/` on kedi (`report.md`, `report_phase1.md`, `metrics/test/`, `selection/`, `listening/`).
- **Code:** `src/steering.py`, `src/snac_encode.py`, `src/caa_stats.py`, `src/caa_pilot.py`, `scripts/caa_extract.py`, `scripts/caa_generate.py`, `scripts/caa_evaluate.py`, `configs/caa_pilot/`, `tests/test_caa.py` (7 tests passing on kedi). Not yet committed. The run's `provenance/` directory records base commit `dd6044b` plus the uncommitted diff and package versions.
- **Compute:** RTX A5500 GPUs on kedi. 600 teacher-forced passes; 624 + 912 dev clips; 1,197 test clips.
- **Evaluation models:** Whisper-large-v3, `microsoft/wavlm-base-plus` (rev. 4c66d480), `microsoft/wavlm-base-plus-sv`. Statistics are implemented in numpy/torch.